Unit: 6 Unit 1: Ratio, Rate, Percentage **Name:** ________________
Topic: Ratios Proportions **Date:** __________

Homework (Jalapeno)

Q1. In ancient Egypt, the ratio of the pyramid's base to its height was 3 : 4. If the height of the pyramid is 80 meters, what is the base?
(A) 60 meters
(B) 90 meters
(C) 120 meters
(D) 100 meters
(E) 40 meters

Q2. A recipe for ancient Greek soup requires a ratio of 2 : 5 for water to vegetables. If 10 liters of water are used, how many kilograms of vegetables are needed?
(A) 25 kg
(B) 20 kg
(C) 15 kg
(D) 10 kg
(E) 5 kg

Q3. The ancient Babylonians used a sexagesimal system with a ratio of 6 : 10 for their number system. If a number in this system is 24, what is its equivalent in the decimal system?
(A) 4
(B) 6
(C) 8
(D) 10
(E) 12

Q4. In ancient China, the ratio of the emperor's palace to the peasant's hut was 10 : 1. If the peasant's hut is 50 square meters, what is the area of the emperor's palace?
(A) 500 square meters
(B) 1000 square meters
(C) 5000 square meters
(D) 10000 square meters
(E) 50000 square meters

Q5. A proportion states that 8 ancient coins are equivalent to 12 modern coins. If 24 ancient coins are used, how many modern coins are needed?
(A) 12
(B) 18
(C) 24
(D) 30
(E) 36

Q6. The ratio of men to women in an ancient Roman village was 3 : 2. If there were 72 men, how many women were there?
(A) 36
(B) 40
(C) 48
(D) 50
(E) 60

Q7. An ancient Mayan artifact has a golden ratio of 1.618 : 1. If the length of the shorter side is 10 cm, what is the length of the longer side?
(A) 15.18 cm
(B) 16.18 cm
(C) 20 cm
(D) 25 cm
(E) 30 cm

Q8. The ratio of an ancient Phoenician ship's length to its width was 5 : 2. If the width is 8 meters, what is the length of the ship?
(A) 10 meters
(B) 16 meters
(C) 20 meters
(D) 25 meters
(E) 30 meters

Open Ended Questions (FRQ)

Q9. The ancient Egyptians built the Great Pyramid with a ratio of 4 : 3 for the base to the height. If the pyramid's base is 150 meters and its original height was 200 meters, what is the new height after 20 meters of erosion? Show your work and provide the new ratio of base to height.

Q10. The ancient Greeks used a ratio of 1 : 1.618 for their architectural designs. If a building's shorter side is 15 meters, what is the length of the longer side? Use this ratio to find the dimensions of the building if the shorter side is increased by 25%. Provide your answer with the new dimensions and the ratio of the longer side to the shorter side.

Chef's Tips

- Tip 1: Make sure to check the ratio and proportion calculations carefully.
- Tip 2: Use cross multiplication to solve proportions.
- Tip 3: Model real-world situations with proportions to ensure understanding.